

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
# Create DataFrame
data = {
    "experience": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "salary" :
        [30000, 35000, 40000, 45000, 52000,
         58000, 65000, 72000, 80000, 90000] }

df = pd.DataFrame(data)
df
```

	experience	salary	
0	1	30000	
1	2	35000	
2	3	40000	
3	4	45000	
4	5	52000	
5	6	58000	
6	7	65000	
7	8	72000	
8	9	80000	
9	10	90000	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
# Features and target
X = df[["experience"]]
y = df["salary"]
y
```

	salary
0	30000
1	35000
2	40000
3	45000
4	52000
5	58000
6	65000
7	72000
8	80000
9	90000

dtype: int64

```
df.shape
df.info()
df.describe()
df.columns
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  -
0    experience   10 non-null     int64
1    salary       10 non-null     int64
dtypes: int64(2)
memory usage: 292.0 bytes
Index(['experience', 'salary'], dtype='object')
```

```
X_train, X_test, y_train, y_test = train_test_split(X,y, test_size= 0.3, random_state = 42)
```

X_train,y_train

```
(   experience
0             1
7             8
2             3
9            10
4             5
3             4
6             7,
0    30000
7    72000
2    40000
9    90000
4    52000
3    45000
6    65000
Name: salary, dtype: int64)
```

```
# Model
model = LinearRegression()
model
```

LinearRegression

i

?

LinearRegression()

#Training

```
model.fit(X_train, y_train)
```

▼ LinearRegression ⓘ ?

LinearRegression()

```
# Coefficients
print("Intercept ( $\beta_0$ ):", model.intercept_)
print("Coefficient ( $\beta_1$ ):", model.coef_[0])
```

```
Intercept ( $\beta_0$ ): 20341.584158415826
Coefficient ( $\beta_1$ ): 6621.287128712874
```

```
# Prediction
y_pred = model.predict(X_test)
y_pred
```

```
array([79933.16831683, 33584.15841584, 60069.30693069])
```

```
# y = b0 + b1(x)
y = 20341 + 6621 * 2
```

```
print(y)
print(X_test)
print(y_test)
```

```
35000 - 33583
```

```
33583
```

```
experience
```

```
8          9
```

```
1          2
```

```
5          6
```

```
8      80000
```

```
1      35000
```

```
5      58000
```

```
Name: salary, dtype: int64
```

```
1417
```

X_test

experience	
8	9
1	2
5	6

Next steps: [Generate code with X_test](#) [New interactive sheet](#)

```
# Metrics
mae = mean_absolute_error(y_test, y_pred)
```

```
mse = mean_squared_error(y_test, y_pred)
```

```
# mae
```

```
r2 = r2_score(y_test, y_pred)
```

```
# print(r2)
```

```
rmse = np.sqrt(mse)
```

```
print("mse:", mse)
```

```
print("mae:", mae)
```

```
print("rmse:", rmse)
```

```
print("r2_score:", r2)
```

```
mse: 2097035.0129072436
mae: 1183.9933993399343
rmse: 1448.1142955261657
r2_score : 0.9937875855443827
```

```
# Metrics
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\nMetrics")
print("MAE :", mae)
print("MSE :", mse)
print("RMSE:", rmse)
print("R² :", r2)
```

```
Metrics
MAE : 1183.9933993399343
MSE : 2097035.0129072436
RMSE: 1448.1142955261657
R² : 0.9937875855443827
```

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)

# Create DataFrame
data = {
    "experience": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "education_years": [12, 13, 14, 15, 16, 16, 17, 18, 18, 19],
    "salary": [30000, 35000, 40000, 45000, 52000,
               58000, 65000, 72000, 80000, 90000]
}

df = pd.DataFrame(data)
df
```

	experience	education_years	salary	
0	1	12	30000	
1	2	13	35000	
2	3	14	40000	
3	4	15	45000	
4	5	16	52000	
5	6	16	58000	
6	7	17	65000	
7	8	18	72000	
8	9	18	80000	
9	10	19	90000	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
# Features and target
X = df[["experience", "education_years"]]
y = df["salary"]
```

```

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Model
model = LinearRegression()

# Training
model.fit(X_train, y_train)

# Intercept
print("Intercept ( $\beta_0$ ):", model.intercept_)

# Coefficients
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_
})

print("\nCoefficients:")
print(coefficients)

# Prediction
y_pred = model.predict(X_test)

# Metrics
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\nMetrics")
print("MAE :", mae)
print("MSE :", mse)
print("RMSE:", rmse)
print("R2 :", r2)

```

Intercept (β_0): 73520.58823529402

Coefficients:

	Feature	Coefficient
0	experience	10114.705882
1	education_years	-4576.470588

Metrics

MAE : 1460.2941176470558
MSE : 2645367.6470588082
RMSE: 1626.4586213792247
R² : 0.9947745824255628

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